

Customer Shopping Behaviour Analysis

An End-to-End Data Analysis Project

Python

MySQL

Power BI

Data-driven insights into customer purchasing patterns



Project Overview



Phase 1 — Python

- Data loading & exploration
- Missing value imputation
- Column standardisation
- Feature engineering
- Export to MySQL



Phase 2 — SQL

- 10 business questions
- Revenue & customer analysis
- Segmentation with CTEs
- Window functions
- Discount & shipping trends



Phase 3 — Power BI

- Interactive dashboard
- Revenue breakdowns
- Customer segmentation
- Product performance
- Subscription insights

Dataset Overview

3,900+

Customer Records

18

Original Features

10

SQL Questions

3

New Features Added

Feature	Description	Feature	Description
customer_id	Unique identifier	subscription_status	Active / Inactive
age / age_group*	Age & derived segment	shipping_type	Standard, Express, etc.
gender	Male / Female	discount_applied	Yes / No
item_purchased	Product name	previous_purchases	Count of past orders
category	Product category	frequency_of_purchases	Weekly, Monthly, etc.
purchase_amount	Value in USD	purchase_freq_days*	Numeric days (derived)
review_rating	1–5 star score	age_group*	Quartile-based segment

** Engineered features added during Python data cleaning*



Phase 1 — Python: Data Cleaning & Feature Engineering

01 Data Exploration

Loaded CSV with pandas. Inspected shape, dtypes, and descriptive stats via `.info()` and `.describe()`.

02 Missing Values

Identified nulls in `review_rating`. Imputed with median per category using `groupby + transform`.

03 Column Standardisation

Converted all column names to lowercase with underscores. Renamed `purchase_amount_(usd)`.

04 Feature Engineering

Created `age_group` (`pd.qcut`, 4 quartiles) and `purchase_frequency_days` (text-to-numeric mapping).

05 Removed Redundancy

Confirmed 100% match between `discount_applied` and `promo_code_used`. Dropped `promo_code_used`.

06 MySQL Export

Connected via SQLAlchemy + PyMySQL. Loaded cleaned DataFrame into `customer_behavior_db`.



Phase 2 — SQL: Business Questions Answered

Q1

Total revenue by gender

GROUP BY / SUM

Q2

Discounted buyers above avg spend

Subquery / WHERE

Q3

Top 5 products by review rating

ORDER BY / LIMIT

Q4

Standard vs Express purchase amounts

Conditional WHERE

Q5

Subscriber vs non-subscriber spend

COUNT / AVG / SUM

Q6

Top 5 products by discount rate

CASE WHEN / %

Q7

Segment: New / Returning / Loyal

CTE / CASE WHEN

Q8

Top 3 products per category

ROW_NUMBER()
Window

Q9

Repeat buyers & subscription link

Filtered GROUP BY

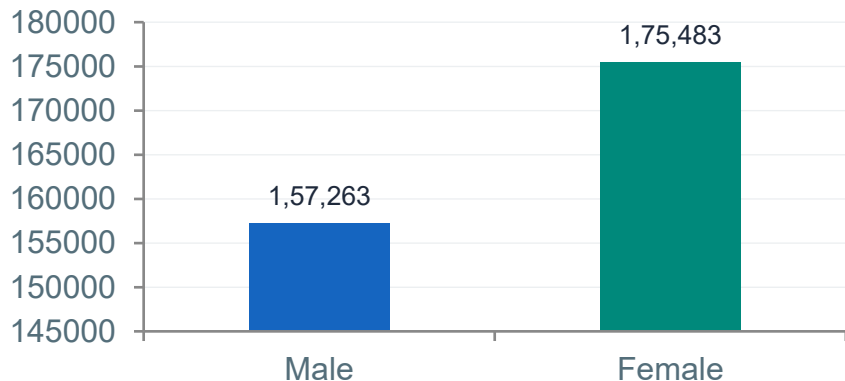
Q10

Revenue by age group

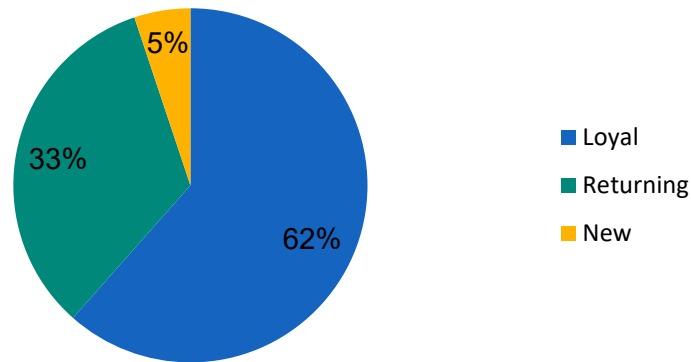
Derived column /
SUM

Key Insights from SQL Analysis

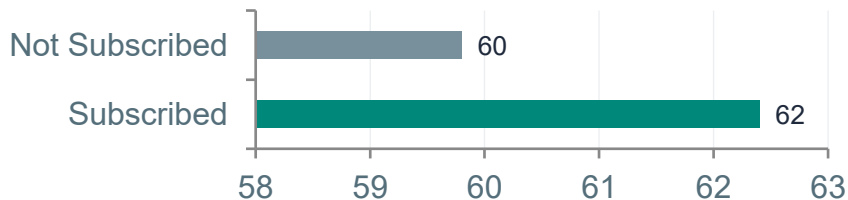
Revenue by Gender



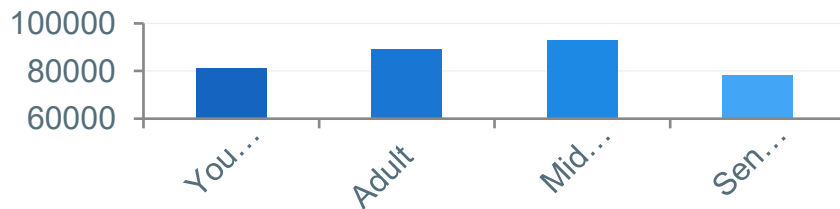
Customer Segments



Avg Spend: Subscribed vs Non-subscribed (\$)

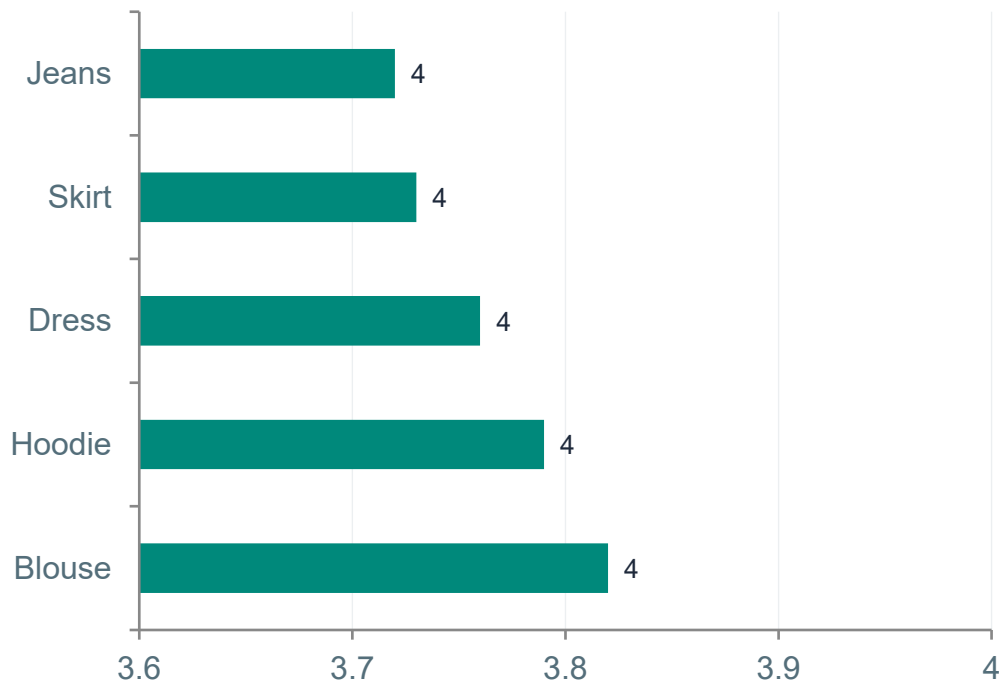


Revenue by Age Group (\$)

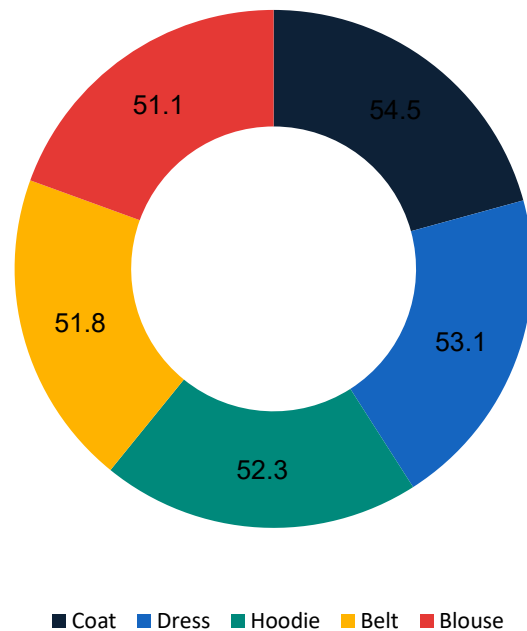


Product & Discount Insights

Top 5 Products — Avg Review Rating



Top 5 — Discount Rate %





Phase 3 — Power BI Interactive Dashboard

Customer Behavior Dashboard

Subscription Status

No Yes

Gender

Female Male

Category

Accessories

Clothing

Footwear

Outerwear

Shipping Type

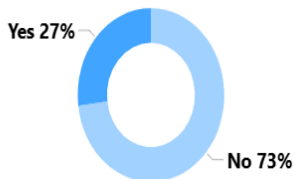
- ☐ 2-Day Shipping
- ☐ Express
- ☐ Free Shipping
- ☐ Next Day Air
- ☐ Standard
- ☐ Store Pickup

3.9K
Number of Customers

\$59.76
Average Purchase Amount

3.75
Average review Rating

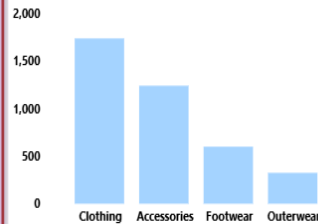
% of Customers by Subscription Status



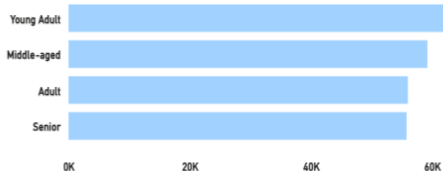
Revenue by Category



Sales by Category



Revenue by Age Group



Sales by Age Group



Dashboard Highlights

- ✓ Multi-page interactive report
- ✓ Slicers: Gender, Age, Category
- ✓ Revenue & spend KPIs
- ✓ Subscription vs non-subscriber view
- ✓ Product & category breakdown
- ✓ Shipping type performance

Key Findings



Female customers drive more revenue

Female customers generated ~\$175K vs ~\$157K from male customers — an 11% difference.



Most customers are Loyal buyers

Over 60% of customers have more than 10 previous purchases, showing strong retention.



Subscribed customers spend more

Subscribers show higher average spend and contribute disproportionately to total revenue.



Apparel dominates high ratings

Blouse, Hoodie, and Dress lead in review ratings, all scoring above 3.75 on average.



Shipping choice doesn't affect spend

Standard and Express shipping users show comparable average purchase amounts.



Middle-aged segment tops revenue

Middle-aged customers contribute the highest total revenue among all age groups.



Thank You

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Python · MySQL · Power BI

Python

pandas

SQLAlchemy

MySQL

Power BI